

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)

Data Visualization with Power BI and Tableau

Module - II

Practical – IX

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Class:TYIT	Batch:2
Date of Assignment: 27/08/26	Date/Time of Submission: 27/08/26

9.Cleaning and Structuring Data for Analysis.

a) Identify and handle missing or inconsistent data.

STEPS:

1.Select and open Sem1_Scores.csv.

2. Right-click an empty space in the Data Pane select Create Calculated Field.

3. Create the Field and named it as Physics_Clean.

Tableau Formula: `FLOAT(REPLACE(STR([Physics]), '%', ''))`

4. Create the Field and named it as Attendance_Clean.

Tableau Formula: `FLOAT(TRIM(REPLACE(STR([Attendance]), '%', '')))`

5. Create the Field and named it as Math_Clean.

Tableau Formula: `IFNULL([Math], 0)`

6. Create the Field and named it as Chemistry_Clean

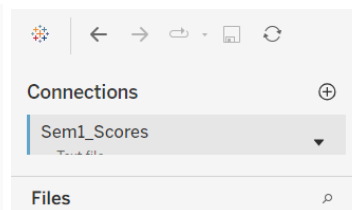
Tableau Formula: `IFNULL([Chemistry], 0)`

7.right-click on Physics_Clean --> Edit:

Tableau Formula: `IFNULL(FLOAT(REPLACE(STR([Physics]), '%', '')), 0)`

Sem1 Scores.csv:

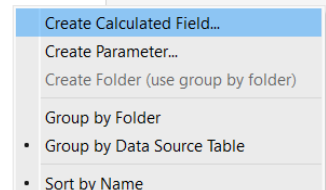
Student_ID,Full_Name,Dept_Info,Math,Physics,Chemistry,Programming,Attendance
S101, Aarav Sharma ,CSE-01,85,90%,88,95,85 %
S102,priya VERMA,ECE-02,,75%,80,85,90 %
S103,Rohan Gupta,CSE-01,78,82%,,88,78 %
S104,Sneha Patel,IT-03,92,88%,90,96,95 %
S105,aditya Rao ,MECH-04,45,,50,60,65 %
S106,Ananya Singh,CSE-01,88,91%,85,92,88 %
S107,Rahul Nair,ECE-02,,70%,75,80,72 %
S108,Kavya Reddy,IT-03,95,94%,92,98,98 %
S109,Vikram Joshi,CSE-01,67,60%,58,72,80 %
S110,Pooja Iyer,ECE-02,82,85%,80,89,91 %



Sem1_Scores

Sem1_Scores.csv

Student ID	Full Name	Dept Info	Math	Physics	Chemistry	Programming	Attendance
S101	Aarav Sharma	CSE-01	85	0.900000	88	95	85
S102	priya VERMA	ECE-02	null	0.750000	80	85	90
S103	Rohan Gupta	CSE-01	78	0.820000	null	88	78
S104	Sneha Patel	IT-03	92	0.880000	90	96	95
S105	aditya Rao	MECH-04	45	null	50	60	65
S106	Ananya Singh	CSE-01	88	0.910000	85	92	88



Physics_Clean

`FLOAT(REPLACE(STR([Physics]), '%', ''))`

Attendance_Clean

`FLOAT(TRIM(REPLACE(STR([Attendance]), '%', '')))`

The calculation is valid.

Apply

OK

The calculation is valid.

Apply

OK

Math_Clean

×

Chemistry_Clean

×

IFNULL([Math], 0) |

IFNULL([Chemistry], 0) |

▶

The calculation is valid.

Apply

OK

The calculation is valid.

Apply

OK

Physics_Clean

×

IFNULL(FLOAT(REPLACE(STR([Physics]), '%', '')), 0)

▶

The calculation is valid.

Apply

OK

b) Convert wide data into tall format.

STEPS:

1. Open the Data Source Grid
2. Click the header of the Math column. Hold down Ctrl and click Physics, Chemistry, and Programming so all four columns are highlighted in blue.
3. Right-click on any of the highlighted column headers. Select Pivot.
4. Rename the Pivoted Fields: Subject and Score.
5. Click the Abc icon and Select Number (decimal) from the dropdown list.
6. create a calculated field inside Sheet 1 named Clean_Score: IFNULL(FLOAT(REPLACE(STR([Score]), '%', '')), 0)

Sem1_Scores.csv

12 fields 10 rows

10 → rows

⚙

▼

Name Sem1_Scores.csv

Description No description available.

Fields

Type	Field Name	Physical Table	Remote Fiel...
Abc	Student ID	Sem1_Scores.csv	Student_ID
Abc	Full Name	Sem1_Scores.csv	Full_Name
Abc	Dept Info	Sem1_Scores.csv	Dept_Info
#	Math	Sem1_Scores.csv	Math
#	Physics	Sem1_Scores.csv	Physics
#	Chemistry	Sem1_Scores.csv	Chemistry
#	Programming	Sem1_Scores.csv	Programming
Abc	Attendance	Sem1_Scores.csv	Attendance

Abc	Abc	Abc	#	#	#	#	Abc
Sem1_Scores.csv	Sem1_Scores.csv	Sem1_Scores.csv	Sem1_Scores.csv	Sem1_Scores.csv	Sem1_Scores.csv	Sem1_Scores.csv	Sem1_Scores.csv
Student ID	Full Name	Dept Info	Math	Physics	Chemistry	Programming	Atte
S101	Aarav Sharma	CSE-01	85	0.900000	88	95	85 %
S102	priya VERMA	ECE-02	null	0.750000	80	85	90 %
S103	Rohan Gupta	CSE-01	78	0.820000	null	88	78 %
S104	Sneha Patel	IT-03	92	0.880000	90	96	95 %
S105	aditya Rao	MECH-04	45	null	50	60	65 %
S106	Ananya Singh	CSE-01	88	0.910000	85	92	88 %
S107	Rahul Nair	ECE-02	null	0.700000	75	80	72 %
S108	Kavya Reddy	IT-03	95	0.940000	92	98	98 %
S109	Vikram Joshi	CSE-01	67	0.600000	58	72	80 %
S110	Pooja Iyer	ECE-02	82	0.850000	80	89	91 %

Rename

Copy Values

Hide

Create Calculated Field...

Pivot

Merge Mismatched Fields

Clean_Score

×

IFNULL(FLOAT(REPLACE(STR([Score]), '%', '')), 0)

▶

Describe

No description available.

Fields

Type

#

Abc

#

#

#

Geographic Role

✓ Number (decimal)
Number (whole)
Date & Time
Date
String
Boolean
Default
Geographic Role

The calculation is valid.

Apply

OK

c) Perform union of multiple files.

STEPS:

- 1. right-click it and select Remove to start with a clean canvas.
- 2. Drag the New Union option onto the empty canvas area.
- 3. Drag Sem1_Scores.csv and Drag Sem2_Scores.csv
- 4. to verify the generated system column: Table Name: Displays "Sem1_Scores.csv" for the first 10 rows and "Sem2_Scores.csv" for the remaining 10 rows.
- 5. select Create Calculated Field. Set the Field Name to: Semester
- 6. Enter the following formula:
IF CONTAINS([Table Name], "Sem1") THEN "Semester 1"
ELSEIF CONTAINS([Table Name], "Sem2") THEN "Semester 2"
ELSE "Other"
END

Sem1_Scores.csv

Open...

Rename

Remove

Field names are in first row

Generate field names automatically

Text File Properties...

Convert to Union...

Sem1_Scores.csv

Union

14 fields 20 rows

Union

Specific (manual)

Wildcard (automatic)

Connection: Sem1_Scores

Sem1_Scores.csv

Sem2_Scores.csv

Tables in union: 2

Apply

OK

Name Union

Description

No description available.

Fields

Type	Field Name	Physical Table	Remote Field...
Abc	Student_ID	Union	Student_ID
Abc	Full_Name	Union	Full_Name
Abc	Dept_Info	Union	Dept_Info

Union

Abc Union	Abc Union	Abc Union	# Union	# Union	# Union	# Union	Abc Union
Student_ID	Full_Name	Dept_Info	Math	Physics	Chemistry	Programming	Attendance
S101	Aarav Sharma	CSE-01	85	0.900000	88	95	85 %
S102	priya VERMA	ECE-02	null	0.750000	80	85	90 %
S103	Rohan Gupta	CSE-01	78	0.820000	null	88	78 %
S104	Sneha Patel	IT-03	92	0.880000	90	96	95 %
S105	aditya Rao	MECH-04	45	null	50	60	65 %
S106	Ananya Singh	CSE-01	88	0.910000	85	92	88 %

Union

14 fields 20 rows

Name Union

Description

No description available.

Fields

Type	Field Name	Physical Table	Remote Field...
Abc	Student_ID	Union	Student_ID
Abc	Full_Name	Union	Full_Name
Abc	Dept_Info	Union	Dept_Info
#	Math	Union	Math
#	Physics	Union	Physics

try	# Union	Abc Union	Abc Union	=# Calculation	=# Calculation	=# Calculation
	Programming	Attendance	Table Name	Physics_Clean	Attendance_Clean	Math_Clean
75	80	72 %	Sem1_Scores.csv	0.700000	72.0000	
92	98	98 %	Sem1_Scores.csv	0.940000	98.0000	9
58	72	80 %	Sem1_Scores.csv	0.600000	80.0000	6
80	89	91 %	Sem1_Scores.csv	0.850000	91.0000	8
90	98	88 %	Sem2_Scores.c...	0.920000	88.0000	8
82	88	92 %	Sem2_Scores.c...	0.780000	92.0000	7
80	90	80 %	Sem2_Scores.c...	0.850000	80.0000	8
94	98	96 %	Sem2_Scores.c...	0.900000	96.0000	9

Semester

IF CONTAINS([Table Name], "Sem1") THEN "Semester 1"
ELSEIF CONTAINS([Table Name], "Sem2") THEN "Semester 2"
ELSE "Other"
END

The calculation is valid.

Apply

OK

d) Execute cross-database joins.

STEPS:

1. In the left sidebar under Files / Connections, find Department_Master.csv.
2. Drag Department_Master.csv onto the canvas and drop it to the right of your Unionbox.
3. If the relationship dialog opens (or click the relationship line):
4. Left Table Field Select Dept_Info and Dept_Code with Operator: =
5. From the left sidebar, drag Student_Activities.csv onto the canvas.
- 6.. Drop it to the right of the Union box.
7. Left Table Field Select Student_ID with Operator: =
8. Click on Sheet 1 In the Data Pane on the far left, you will now see all three tables organized with their respective fields:
Union: Student_ID, Full_Name, Dept_Info, Math, Physics, Chemistry, Programming, Attendance, Table Name
Department_Master: Dept_Code, Department_Name, HOD_Name, Dept_Target_Average, Total_Dept_Capacity
Student_Activities: Club_Membership, Projects_Completed, Certifications

Sem1_Scores

Union

Department_Master.csv

Union — Departme...

4 rows

Name Department_Master.csv

How do relationships differ from joins? [Learn more](#)

Union

Operator

Department_Master...

Abc Dept_Info

=

Abc Dept Code

+

 Add more fields

>

 Performance Options

Remove Relationship

Abc Department_Master.csv	Abc Department_Master.csv	Abc Department_Master.csv	# Department_Master.csv	# Department_Master.csv
Dept Code	Department Name	HOD Name	Dept Target Average	Total Dept Capacity
CSE-01	Computer Science	Dr. Raman	85	120
ECE-02	Electronics & Comm	Dr. Iyer	78	90
IT-03	Information Tech	Dr. Sen	82	60
MECH-04	Mechanical Engg	Dr. Sharma	70	60

Union

Department_Master.csv

Student_Activities.csv

Union — Student_A...

Name Student_Activities.csv

How do relationships differ from joins? [Learn more](#)

Union

Operator

Student_Activities....

Abc Student_ID

=

Abc Student ID (Stuc

+

 Add more fields

>

 Performance Options

Remove Relationship

Union

Attendance

Dept_Info

Full_Name

Semester

Student_ID

Table Name

Attendance_Clean

Chemistry

Chemistry_Clean

Math

Math_Clean

Physics

Physics_Clean

Programming

Union (Count)

Measure Names

e) Restructure data for better visualization.

STEPS:

1. Create Calculated Field name it as : Total_Score

Formula: [Math_Clean] + [Physics_Clean] + [Chemistry_Clean] + [Programming]

2. Create Calculated Field name it as : Overall_Percentage

Formula: [Total_Score] / 400.0

3. right-click Overall_Percentage => Default Properties => Number Format. Select Percentage & set decimal place 1

4. Create Calculated Field name it as : Performance_Band

Formula: IF [Overall_Percentage] >= 0.85 THEN "Distinction"

ELSEIF [Overall_Percentage] >= 0.75 THEN "First Class"

ELSEIF [Overall_Percentage] >= 0.60 THEN "Second Class"

ELSE "Needs Academic Support" END

5. Create Calculated Field name it as : Attendance_Status

Formula: IF [Attendance_Clean] >= 75 THEN "Eligible (Good Standing)" ELSE "Shortage Warning (<75%)" END

6. Drag Student_ID & Full_Name onto the Rows and Semester onto the Columns.

3. Drag Overall_Percentage to the Text. Drag Performance_Band onto the Color and Attendance_Status to the Tooltip.

The screenshot displays the Tableau Desktop interface with four calculated fields being created or edited:

- Total_Score:** Formula: `[Math_Clean] + [Chemistry_Clean] + [Physics_Clean] + [Programming]`
- Overall_Percentage:** Formula: `[Total_Score] / 400.0`. A context menu is open over this field, showing options like 'Default Properties', 'Geographic Role', 'Group by', 'Folders', 'Replace References...', and 'Describe...'. The 'Default Properties' submenu is also visible, showing 'Comment...', 'Color...', 'Number Format...', 'Aggregation', and 'Total using'. The 'Number Format...' option is selected, opening the 'Default Number Format [Overall_Percentage]' dialog. In this dialog, 'Percentage' is selected under the 'Number' category, and 'Decimal places' is set to 1.
- Performance_Band:** Formula: `IF [Overall_Percentage] >= 0.85 THEN "Distinction" ELSEIF [Overall_Percentage] >= 0.75 THEN "First Class" ELSEIF [Overall_Percentage] >= 0.60 THEN "Second Class" ELSE "Needs Academic Support" END`
- Attendance_Status:** Formula: `IF [Attendance_Clean] >= 75 THEN "Eligible (Good Standing)" ELSE "Shortage Warning (<75%)" END`

Each field editor shows a 'The calculation is valid.' message with 'Apply' and 'OK' buttons. The 'Attendance_Status' field editor also includes a 'Clear' button.

Pages

Filters

Marks

Automatic

Color

Size

Text

Detail

Tooltip

Performance_...

SUM(Overall_P..

Attendance_St..

Columns

Student_ID

Full_Name

Rows

Semester

T054

	Student_ID / Full_Name											
	S101	S102	S103	S104	S105	S106	S107	S108	S109	S110		
Semester	Aarav..	Priya .. priya..	Roha..	Sneha..	Adity.. adity..	Anan..	Rahul ..	Kavya..	Vikra..	Pooja ..		
Semester 1	67.2%	41.4%	41.7%	69.7%	38.8%	66.5%	38.9%	71.5%	49.4%	63.0%		
Semester 2	69.2%	62.4%	63.2%	72.0%	43.1%	68.5%	58.7%	72.7%	51.9%	65.5%		

f) Handle different levels of detail in datasets.

STEPS:

1. Create Calculated Field name it as : Dept_Average_Percentage_LOD
Formula: { FIXED [Dept_Info] : AVG([Overall_Percentage]) }
2. Right-click Dept_Average_Percentage_LOD=>Default Properties=>Number Format=>select Percentage 1
3. Create Calculated Field name it as : Student_Overall_Career_Average
Formula: { FIXED [Student_ID] : AVG([Overall_Percentage]) }
4. Create Calculated Field name it as : Variance_From_Dept_Average
Formula: [Overall_Percentage] - [Dept_Average_Percentage_LOD]
5. Create Calculated Field name it as : Benchmark_Status
Formula: IF [Variance_From_Dept_Average] >= 0 THEN "Above Department Benchmark"
ELSE "Below Department Benchmark" END
6. Drag Dept_Info, Student_ID,& Full_Name to the Rows and Semester to the Columns.
7. Drag Overall_Percentage to Text and Dept_Average_Percentage_LOD to Tooltip and Drag Benchmark_Status to Color.

Dept_Average_Percentage_LOD

{ FIXED [Dept_Info] : AVG([Overall_Percentage]) }

Student_ID

Table Name

Measure Names

Attendance_Clean

Chemistry

Chemistry_Clean

Clean_Score

Dept_Average_Percentage...

Default Properties

Geographic Role

Group by

Folders

Replace References...

Describe...

Comment...

Color...

Number Format...

Aggregation

Total using

The calculation is valid.

Apply

OK

Default Number Format [Dept_Average_Percentage_LOD]

Automatic

Number (Standard)

Number (Custom)

Currency (Standard)

Currency (Custom)

Scientific

Percentage

Custom

Percentage

Decimal places:

1

Clear

OK

Cancel

Overall_Career_Average

{ FIXED [Student_ID] : AVG([Overall_Percentage]) }

The calculation is valid.

Apply

OK

Benchmark_Status

IF [Variance_From_Dept_Average] >= 0 THEN "Above Department Benchmark"

ELSE "Below Department Benchmark"

END

The calculation is valid.

Apply

OK

Variance_From_Dept_Average

[Overall_Percentage] - [Dept_Average_Percentage_LOD]

The calculation is valid.

Apply

OK

Pages	iii Columns	Semester
	Rows	Dept_Info Full_Name Student_ID
Filters	T054	
Marks		
Automatic		
Color	Size	Text
Detail	Tooltip	
Benchmark_St..		
SUM(Dept_Ave..		
SUM(Overall_P..		

Dept_Info	Full_Name	Student..	Seme..	Seme..
CSE-01	Aarav Sharma	S101	67.2%	69.2%
	Ananya Singh	S106	66.5%	68.5%
	Rohan Gupta	S103	41.7%	63.2%
	Vikram Joshi	S109	49.4%	51.9%
ECE-02	Pooja Iyer	S110	63.0%	65.5%
	Priya Verma	S102		62.4%
	priya VERMA	S102	41.4%	
	Rahul Nair	S107	38.9%	58.7%
IT-03	Kavya Reddy	S108	71.5%	72.7%
	Sneha Patel	S104	69.7%	72.0%
MECH-04	Aditya Rao	S105		43.1%
	aditya Rao	S105	38.8%	

g) Apply advanced data cleaning techniques.

STEPS:

1. Create Calculated Field name it as : Full_Name_Clean

Formula: UPPER(LEFT(TRIM([Full_Name]), 1)) +

LOWER(MID(TRIM([Full_Name]), 2, FIND(TRIM([Full_Name]), " " - 1)) + " "

+ UPPER(MID(TRIM([Full_Name]),FIND(TRIM([Full_Name]), " ") + 1, 1)) + LOWER(MID(TRIM([Full_Name]), FIND(TRIM([Full_Name]), " ") + 2, LEN(TRIM([Full_Name]))))

2. Create Calculated Field name it as : First_Name

Formula: TRIM(SPLIT([Full_Name], " ", 1))

3. Create Calculated Field name it as : Last_Name

Formula: TRIM(SPLIT([Full_Name], " ", -1))

4. Create Calculated Field name it as : Dept_Branch

Formula: SPLIT([Dept_Info], "-", 1)

5. Create Calculated Field name it as : Section_ID

Formula: SPLIT([Dept_Info], "-", 2)

6. Create Calculated Field name it as : Student_Display_Label

Formula: [Student_ID] + " - " + [Full_Name_Clean] + " (" + [Dept_Branch] + ")"

7. Place Student_Display_Label on the Rows and Semester on the Columns.

8. Place Overall_Percentage on Text. Performance_Band on Color. Dept_Branch and Section_ID in the Filters.

Full_Name_Clean

UPPER(LEFT(TRIM([Full_Name]), 1)) +

LOWER(MID(TRIM([Full_Name]), 2, FIND(TRIM([Full_Name]), " " - 1)) + " "

UPPER(MID(TRIM([Full_Name]), FIND(TRIM([Full_Name]), " ") + 1, 1)) +

LOWER(MID(TRIM([Full_Name]), FIND(TRIM([Full_Name]), " ") + 2, LEN(TRIM([Full_Name]))))

The calculation is valid.

Apply

OK

First_Name

TRIM(SPLIT([Full_Name], " ", 1))

The calculation is valid.

Apply

OK

Last_Name

TRIM(SPLIT([Full_Name], " ", -1))

The calculation is valid.

Apply

OK

Dept_Branch

SPLIT([Dept_Info], "-", 1)

The calculation is valid.

Apply

OK

The calculation is valid.

Apply

OK

The calculation is valid.

Apply

OK

Section_ID

×

Student_Display_Label

×

SPLIT([Dept_Info], "-", 2)

[Student_ID]+"-"+[Full_Name_Clean]+"("+[Dept_Branch]+")"

The calculation is valid.

Apply

OK

The calculation is valid.

Apply

OK

Pages

Filters

Marks

iii Columns

Rows

Semester

Student_Display_La..

T054

Semester

Student_Display_La..	Seme..	Seme..
S104-Sneha Patel(IT)	69.7%	72.0%
S105-Aditya Rao(M..	38.8%	43.1%
S108-Kavya Reddy(I..	71.5%	72.7%

Automatic

Color

Size

Text

Detail

Tooltip

Performance_..

SUM(Overall_P..

CONCLUSION:

The dataset was successfully cleaned and structured for analysis using Tableau. Missing and inconsistent data were handled, data was reshaped into a suitable format, multiple files were combined using union, and joins were performed where required. Different levels of detail were managed effectively, and advanced data-cleaning techniques were applied to prepare the data for accurate visualization and analysis.